

# Shortest path algorithm

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*Practice quiz*

The shortest path algorithm, to calculate the shortest paths from node  $o$  to all the other nodes in the network, is described as follows.

1. Initialization:  $\mathcal{S} \leftarrow \{o\}$ ,  $\lambda_o = 0$ ,  $\lambda_j = +\infty$ .
2. At each iteration:
  - Select  $i$  in  $\mathcal{S}$ .
  - For each  $(i, j)$ :
    - If  $\lambda_j > \lambda_i + c_{ij}$ ,
      - \*  $\lambda_j \leftarrow \lambda_i + c_{ij}$ .
      - \* If  $\lambda_j < 0$  and  $\lambda_j < (m - 1) \min_{(i,j) \in \mathcal{A}} c_{ij}$ : STOP.
      - \*  $\mathcal{S} \leftarrow \mathcal{S} \cup \{j\}$ .
  - $\mathcal{S} \leftarrow \mathcal{S} \setminus \{i\}$ .
  - If  $\mathcal{S} = \emptyset$ : STOP.
  - Otherwise, start again.

Apply the algorithm to identify the shortest paths from node 1 to all other nodes in the following network, and identify the shortest path tree.

